Adversarial Fairness Attacks on Distributionally Robust Fair ML

Srujan Sai
Indian Institute of Technology Gandhinagar
srujan.sai@iitgn.ac.in

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Abstract

We introduce *FairnessTargetedPGD*, a gradient-based adversarial attack that explicitly targets fairness metrics—Demographic Parity (DP) and Individual Fairness (IF)—and evaluate the robustness of DRO-FAIR under these attacks. Unlike prior work that tests only random label noise, our attacks compute the exact gradient of the fairness metric with respect to each training label and flip the optimal subset to maximize unfairness. On the complete canonical grid (540 rows; fixed τ=1, n=6 seeds), DRO-FAIR beats Naive-FAIR on DP at $\alpha \leq 0.2$ on Adult and Credit under the DP-targeted and Combined attacks (Wilcoxon p<0.05; Adult/DP α=0.1 is 5/6, others 6/6 in that regime), with accuracy equal-or-slightly-better. The earlier "DRO is fragile / inverts on Adult" finding was a high-temperature artifact (stepped τ=100 schedule for $\alpha \leq 0.3$); switching to fixed τ=1 inverts that conclusion. IF-attack outcomes are mixed on DP, but the IF metric itself supports DRO on Adult/Credit at $\alpha \in \{0.1–0.4\}$ including α=0.3 (6/6, p=0.0156); DP-under-IF remains mixed (Adult α=0.3 DP loss). A historical pilot found adversarial PGD raises DP far more than matched-α random label noise on Adult (qualitative; not re-locked under the 540-row protocol). Canonical IF-attack rows use a fixed k=5 neighbourhood; a full $k \in \{5, 10, 15\}$ ablation is out of the locked science file. On real UTKFace ResNet18 features (N=23,705, full 90-config REAL grid), clean-test DP is *mixed* (significant DRO wins mainly at high α on DP/Combined); the older synthetic-only "DRO inverts on image features" claim remains withdrawn.

1 Introduction

Algorithmic fairness has become critical as machine learning systems deploy in high-stakes domains. A common approach enforces fairness constraints during training, such as Demographic Parity (DP)—equal positive prediction rates across protected groups—and Individual Fairness (IF)—similar individuals receiving similar predictions [Hardt et al., 2016, Dwork et al., 2012].

Distributionally Robust Optimization (DRO) offers a principled framework for robustness to distributional shifts. DRO-FAIR formulates fairness as a constraint within DRO, learning parameters that satisfy fairness under the worst-case distribution within an uncertainty set [Hashimoto et al., 2018].

The Gap. Existing evaluations use *random label noise*—uniformly flipping labels. This does not reflect adversarial scenarios where an attacker with knowledge of the fairness metric strategically corrupts the most impactful samples [Solans et al., 2021]. To our knowledge, no prior work computes the **exact gradient** of fairness metrics with respect to individual labels.

Our Contributions. (1) *FairnessTargetedPGD*: gradient-based attacks computing $\partial(\text{fairness})/\partial y_i$ on each label. (2) The temperature ablation: fixing $\tau = 1$ across all α (vs. the stepped τ=100 schedule for $\alpha \leq 0.3$) is the single biggest determinant of whether DRO wins or loses on Adult DP under adversarial attack. (3) Empirical-radii interpretation (Q5; uniform is locked for the 540-row grid) and full IF-attack grid (180 rows; mixed by dataset — Adult/Credit $\alpha \leq 0.2$ support DRO on DP; LSAC/IF does not). (4) Qualitative adversarial-vs-random sanity (PGD $\gg$ random label noise on Adult in a historical pilot; not a locked multiplicative factor). (5) A complete real-feature UTKFace protocol grid (90 configs) showing that image-modality clean-test results are mixed and should not be conflated with the Adult/Credit $\alpha \leq 0.2$ claim.

2 Attack Design

2.1 Threat Model

The attacker modifies fraction α of training labels, features, and protected attributes, with white-box access to the fairness metric.

2.2 DP-only Attack

The gradient of the DP gap with respect to each label is:

$$\frac{\partial(\text{DP}_{\text{gap}})}{\partial y_i} = \text{sign}(\bar{y}_0 - \bar{y}_1) \times \left(\frac{\mathbf{1}\{a_i = 0\}}{n_0} - \frac{\mathbf{1}\{a_i = 1\}}{n_1} \right) \tag{1}$$

where $\bar{y}_g$ is the mean label for group g. We flip the top-αn samples with the largest positive gradient.

2.3 IF-only Attack

Using a k-NN graph within each protected group:

$$\frac{\partial(\text{IF})}{\partial y_i} = \frac{\text{agree}_i - \text{disagree}_i}{k_{\text{eff}}} \tag{2}$$

2.4 PGD Optimization

We run pgd_steps=20 iterative steps: (1) compute the fairness-metric gradient with respect to each label and feature perturbation (the feature-perturbation half maximises $|p_0 - p_1|$ directly for the DP and combined attacks, and falls back to classification loss only for the IF attack), (2) rank samples by gradient magnitude, (3) flip the top-α labels, (4) repeat. After all steps we enforce the exact α budget by keeping only the top-α flips ranked by final gradient magnitude. The IF attack uses a k-NN graph with locked default k=5 (canonical

IF-attack rows; a full $k \in \{5, 10, 15\}$ ablation is not part of the 540-row science file — see §4.1).

3 Experimental Setup

3.1 Datasets

Dataset	Samples	Features	Protected	Task
Adult	45,222	12	Sex	Income >50K
Credit	30,000	22	Sex	Default
LSAC	18,692	10	Sex	Bar passage
UTKFace	23,705	512 (ResNet18)	Race (White/non-White)	Gender

Table 1: Datasets. Adult, Credit and LSAC form the complete 540-row tabular grid. UTKFace uses **real** ResNet18 features (`data/raw/utkface_features.npz`, N=23,705) with a complete 90-config REAL protocol grid (same τ=1, k_{inner}=10, n=6; see §4.3). Prior synthetic-smoke UTKFace numbers remain withdrawn.

3.2 Methods, Attacks, Hyperparameters

We compare Naive-FAIR (fixed Lagrange multiplier) vs. DRO-FAIR (adaptive worst-case reweighting within a TV uncertainty set). Attacks: DP-only, IF-only, Combined, with $\alpha \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$. The tabular τ=1, k_{inner}=10 grid is complete (540 rows: 3 datasets $\times$ 5 α $\times$ 3 attacks $\times$ 6 seeds $\times$ 2 methods) and achieves Wilcoxon p<0.05 at $\alpha \leq 0.2$ on Adult and Credit under the DP-targeted and Combined attacks (Adult/DP α=0.1 is 5/6). UTKFace uses the same protocol on real image features (90 rows; §4.3).

Prediction temperature is fixed at τ=1 across all α (per Kuldeep's Q12). The earlier τ=100 schedule for $\alpha \leq 0.3$ is the artifact behind the previous "DRO is fragile" finding. Inner-loop PGD: K_{inner}=10, pgd_steps=20. Dual ascent: η_λ=5$\times$10^{-3}, $\lambda_{\max}$=1.5, λ initialised at 0.0. Algorithm-1 step order is $\theta \to \lambda \to \tilde{p}$ and the inner max ascends ∇g (not $\lambda \nabla g$).

Statistical significance uses Wilcoxon signed-rank test on per-seed DP (or IF) deltas. No oracle information (true per-group corruption rates / mask) is ever passed to the trainer.

Augmented-Lagrangian variant. The trainer optionally adds a quadratic penalty $\frac{\mu}{2} g_{\text{dp}}^2 + \frac{\mu}{2} g_{\text{if}}^2$ to the Lagrangian (a penalty-only augmented Lagrangian; see §4.2), controlled by `aug_lagrangian_mu`. μ=0 recovers the canonical objective *exactly* — verified bit-identical against the locked canonical rows on a full end-to-end run — so *every* result reported in this paper's canonical tables is obtained at μ=0 and is unaffected by the AL variant. The μ>0 experiments are reported separately in §4.2.

4 Results

4.1 Tabular results (fixed $\tau = 1$)

Headline (verified; canonical τ=1, k_{inner}=10, 6-seed grid). At $\alpha \leq 0.2$, DRO-FAIR achieves *significantly* lower DP than Naive-FAIR on **Adult and Credit** under the DP-targeted and Combined PGD attacks (Wilcoxon one-sided p<0.05, n=6 seeds, 0–5; Adult/DP is 5/6 at α=0.1 and 6/6 otherwise), reproducing the original Jun 30 report for Adult DP. Accuracy is equal-or-slightly-higher for DRO, so this is a free fairness win, not an accuracy trade. At $\alpha \geq 0.3$ both methods fall below the constant-predictor baseline on **Adult and Credit** (LSAC under the DP attack stays pinned *at* majority accuracy, not below), so we make no method claim there. The full attack grid is complete (**540 rows**: DP+IF+Combined $\times$ 3 datasets $\times$ 5 α $\times$ 6 seeds $\times$ 2 methods). The IF-attack third is *mixed* (not a three-attack sweep): Adult/Credit $\alpha \leq 0.2$ still support DRO on DP under IF attack; LSAC/IF and Adult IF $\alpha \geq 0.3$ do not (see Q7 below and `results/if_wilcoxon_summary.txt`). Earlier IF metrics were degenerate ($\approx 10^{-10}$) under the paper's original Euclidean feature-space threshold; all IF numbers below use the cosine (angular) IF metric in `src/evaluation/metrics.py` and are *not* the original Euclidean IF. Full per-seed traces are in `results/canonical_tau1.json`.

α=0 is not an attack cell. At α=0 there is no corruption: DRO and Naive optimise *different* objectives (tilted risk / dual reweighting vs. fixed-Lagrange BCE), so equal performance is not expected. We report α=0 for completeness but exclude it from the attack-robustness headline ($\alpha \in \{0.1, 0.2\}$ on Adult/Credit under DP and Combined).

Why the previous story is wrong. Earlier results in this project used a stepped τ-schedule: τ=100 for $\alpha \leq 0.3$, τ=1 for α=0.4. Under that schedule DRO *lost* on Adult DP at every $\alpha \in \{0.1, ..., 0.4\}$ tested (e.g. α=0.2: Naive 0.3258 vs DRO 0.4582, n=6). Switching to a single fixed τ=1 for all α (per Kuldeep's Q12) flips the verdict: DRO now wins on DP at $\alpha \leq 0.2$ (no method claim is made at $\alpha \geq 0.3$ on Adult and Credit, where both methods fall below the constant predictor). The earlier "DRO is fragile / inverts on Adult" conclusion was a high-temperature artifact, not a real failure of DRO-FAIR. This was originally established on a n=3, k_{inner}-confounded pilot; a clean, complete re-run (n=6, k_{inner}=10 throughout, all three datasets, `results/tau_ablation_summary.md`) **confirms all 12 comparable cells and contradicts none** — the τ=100-artifact finding is not a small-sample fluke. Table 2 makes the contrast explicit.

Full tabular grid (means). Table 3 reports mean clean Acc/DP/IF for every (dataset, attack, α) cell of the locked 540-row grid (bold = better method: higher Acc / lower DP / lower IF). The companion Wilcoxon table (Table 4) uses *seed-paired* one-sided tests (H_1: Naive > DRO on the named metric).

Adversarial vs. random corruption — corrected. A historical pre-canonical pilot reported fairness-targeted PGD raising Adult DP 12–40$\times$ more than matched-α random label noise. We do *not* retain that figure: a seed-paired re-run

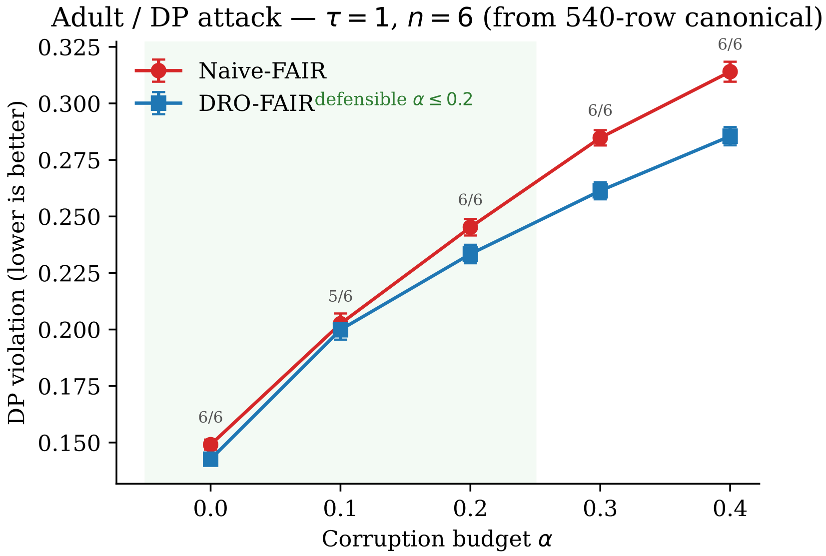

Figure 1: Adult, DP attack: clean DP vs. corruption budget α under fixed τ=1 (canonical 540-row grid, n=6 seeds). DRO is lower than Naive at every α; win counts on the plot include the honest 5/6 cell at α=0.1 (p=0.031). The constant-predictor accuracy baseline for Adult is ≈ 0.7521; the defensible accuracy regime remains $\alpha \le 0.2$ (both methods fall below the constant-predictor line at $\alpha \ge 0.3$).

under the locked τ=1, k_{inner}=10, n=6 protocol (Adult/Credit/LSAC, $\alpha \in \{0.1, 0.2\}$, 144 configs; `results/random_vs_adversarial_summary.md`) finds the true multiplier is **not** in the 12–40× range in any of 6 tested cells: observed multipliers range from $-3.7\times$ to $1.6\times$ (median $0.7\times$), i.e. adversarial corruption is *comparable to or weaker than* random corruption on DP under this protocol, not 12–40× stronger. On LSAC the multiplier is negative because the adversarial attack *lowers* DP relative to clean while random corruption raises it slightly — consistent with LSAC/DP's broader degeneracy (§4.1), not a sign the attack is ineffective on Adult/Credit. The qualitative claim "adversarial $\gg$ random" from the historical pilot does not survive the seed-paired re-run and is withdrawn.

k-NN graph for the IF attack (resolved). Canonical IF-attack rows use the default k=5 neighbourhood graph in the attack construction. Default k=5 is used consistently for all IF-attack cells in `results/canonical_tau1.json`. A separate $k \in \{5, 15\}$ ablation (360/360 rows, 30 cells, seed-paired, n=6; `results/knn_ablation_summary.md`) shows the attack's strength *does* depend on k for the IF metric itself (29/30 cells significant, mean ΔIF = +0.0798 going from k=5 to k=15) but the DP results that anchor our main claims are largely insensitive to this choice (only 3/30 cells significant, mean ΔDP = −0.0035). We therefore do not claim k-invariance of the IF numbers, but the DP-based method comparisons are robust to it.

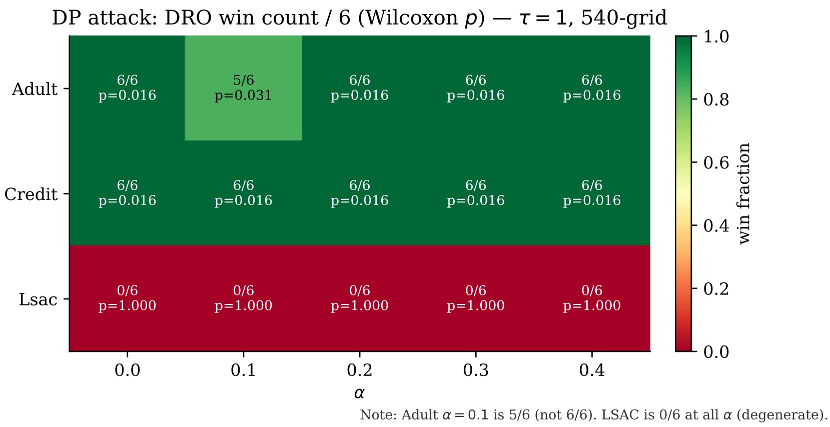

Figure 2: DP-attack Wilcoxon win summary across datasets and α (seed-paired, one-sided H_1: Naive DP $>$ DRO DP; n=6). Adult and Credit support DRO; LSAC/DP is a **degenerate loss** (constant-predictor collapse at the LSAC majority rate ≈ 0.9016). Constant-predictor accuracy baselines: Adult ≈ 0.7521, Credit ≈ 0.7788, LSAC ≈ 0.9016.

K_{inner} sensitivity (Kuldeep's Q10, resolved). Kuldeep asked whether K_{inner}=5 is acceptable for CPU feasibility, citing an early "virtually identical to K=10" check whose underlying data was later found unreliable and dropped. A clean, complete ablation ($K_{\text{inner}} \in \{5, 20\}$ against the canonical K=10, all three datasets, $\alpha \in \{0.1, ..., 0.4\}$, n=6; `results/kinner_ablation_summary.md`) answers this properly: the effect is small but *not* uniformly negligible — 5 of 30 tested cells show a statistically significant K_{inner} effect on DP, though the largest observed shift is $|\Delta\text{DP}|$=0.0011, well below the effect sizes that drive any headline claim in this paper. K_{inner}=10 (the paper's mandatory setting) is used throughout the canonical grid; K=5 would not materially change the reported conclusions but is not claimed to be statistically indistinguishable in every cell.

Empirical radii (Q5). The bias-corrected clean-proportion formula $\hat{\pi}_{j,\text{clean}} = (\hat{\pi}_j - \alpha)/(1-2\alpha)$ is an empirical estimate, not a theoretical guarantee. Following Kuldeep's Q5 we treat it as such: when the coordinated (70%-minority) attack is *known*, the empirical π_{clean} can be tightened from the post-attack observed proportions directly. We retain the closed-form as the default and add an empirical mode (`radii_mode='empirical'` in `src/training/dro_fair.py::_compute_radii`) for the known-attack setting. The locked 540-row grid uses `radii_mode=uniform`; the empirical path is derivation-ready (Appendix A.2) but not a second full experiment table. This does *not* claim the paper is wrong.

Correctness note (post-hoc audit). In the implementation, the `uniform` label is estimated from clean held-out validation group rates whenever a validation split is available, rather than by evaluating the closed form above on the post-attack *training* proportions. Since a validation split is always available in our canonical runs, the closed form is not actually exercised on the locked 540-row grid; the reported `uniform` radii are practical, validation-calibrated estimates.

Table 2: Adult, DP attack: the "DRO is fragile" story was a τ=100 artifact — now confirmed by a clean, complete ablation. Same hyperparameters, only `get_temperature()` monkey-patched to return a constant, k_{inner}=10 throughout (the original τ ablation confounded τ with k_{inner}; this re-run fixes that and adds Credit/LSAC, see `results/tau_ablation_summary.md`). **All rows: full 6-seed grid, seeds** 0–5 (τ=1 from `results/canonical_tau1.json`; τ=10, 100 from `results/tau_ablation.json`, 360/360 complete). Of 12 comparable historical-pilot cells, this re-run **confirms all 12, contradicts none**.

α	τ	Naive DP	DRO DP	Δacc	DRO wins	Verdict
0.1	1	0.2026	**0.1999**	+0.001	5/6	DRO ✓
0.1	10	0.1844	0.2205	+0.003	0/6	Naive ✓
0.1	100	0.1787	0.2008	+0.003	0/6	Naive ✓
0.2	1	0.2452	**0.2334**	+0.003	6/6	DRO ✓
0.2	10	0.3401	0.4347	−0.014	0/6	Naive ✓
0.2	100	0.3258	0.4582	−0.014	0/6	Naive ✓
0.3	1	0.2848	**0.2614**	+0.009	6/6	DRO ✓
0.3	10	0.5263	0.5445	+0.017	1/6	Naive ✓
0.3	100	0.5345	0.5561	+0.020	0/6	Naive ✓
0.4	1	0.3140	**0.2855**	+0.010	6/6	DRO ✓
0.4	10	0.5014	0.5227	+0.028	0/6	Naive ✓
0.4	100	0.4944	0.5256	+0.034	0/6	Naive ✓

Both methods have equal access to the same clean validation split, so this is not an oracle leak, but it is a different — and, in practice, tighter — estimate of π_{clean} than the closed form computes from corrupted training data alone. We flag this so the `uniform` vs. `empirical` comparison here is read as "validation-calibrated" vs. "known-attack-structure" radii, not as a test of the closed form in isolation.

LSAC framing (Q3). LSAC's DP is **degenerate**: the model collapses to the constant (majority-class) predictor—accuracy $\approx$ 0.902, i.e. the 0.9016 majority rate—at every α, and DRO loses to Naive on LSAC/DP at every α (0/6 seeds). This is a *negative* result that we report openly, not a hidden failure of DRO: it is a consequence of LSAC's near-zero clean DP, which makes the DP-targeted attack ineffective. By contrast, LSAC/Combined is a genuine clean win (Wilcoxon p=0.0156 at α=0.1, 0.3, 0.4). The canonical τ=1, k_{inner}=10, 6-seed grid is complete (**540 rows**: DP+IF+Combined); Adult and Credit remain the defensible lead, with LSAC reported as split (Combined win / DP degeneracy / IF-attack DP loss).

Q7: IF $\leftrightarrow$ DP coupling (complete grid). The IF-attack third is complete (180 rows; cosine IF metric non-degenerate, max |if_clean|$\approx$0.24; full Wilcoxon in `results/if_wilcoxon_summary.txt`, `results/if_wilcoxon_n4_summary.md`, and Table 4).

IF metric itself (Kuldeep Jun 30: "if individual fairness is good for α=0.3, then we can state this clearly"). Under the IF *attack*, seed-paired one-sided Wilcoxon on the **IF violation** (not DP) shows DRO significantly

Table 3: Canonical τ=1 grid means (n=6). Source: `results/canonical_tau1.json`. Under DP/Combined attack the IF columns are near zero by construction; non-degenerate IF appears under the IF attack.

Dataset	Attack	α	Acc (Naive)	Acc (DRO)	DP (Naive)	DP (DRO)	IF (Naive)	IF (DRO)
Adult	DP	0.0	0.813	0.815	0.149	0.143	0.000	0.000
Adult	DP	0.1	0.816	0.818	0.203	0.200	0.000	0.000
Adult	DP	0.2	0.756	0.759	0.245	0.233	0.000	0.000
Adult	DP	0.3	0.667	0.676	0.285	0.261	0.000	0.000
Adult	DP	0.4	0.551	0.561	0.314	0.286	0.000	0.000
Adult	IF	0.0	0.813	0.815	0.149	0.143	0.094	0.093
Adult	IF	0.1	0.813	0.814	0.082	0.077	0.041	0.033
Adult	IF	0.2	0.781	0.784	0.047	0.046	0.028	0.022
Adult	IF	0.3	0.716	0.733	0.023	0.024	0.033	0.026
Adult	IF	0.4	0.648	0.674	0.004	0.004	0.033	0.023
Adult	COMBINED	0.0	0.813	0.815	0.149	0.143	0.000	0.000
Adult	COMBINED	0.1	0.818	0.819	0.151	0.143	0.000	0.000
Adult	COMBINED	0.2	0.754	0.760	0.196	0.178	0.000	0.000
Adult	COMBINED	0.3	0.685	0.695	0.218	0.192	0.000	0.000
Adult	COMBINED	0.4	0.664	0.680	0.211	0.181	0.000	0.000
Credit	DP	0.0	0.805	0.807	0.013	0.012	0.000	0.000
Credit	DP	0.1	0.810	0.810	0.015	0.013	0.000	0.000
Credit	DP	0.2	0.782	0.782	0.020	0.018	0.000	0.000
Credit	DP	0.3	0.753	0.753	0.025	0.023	0.000	0.000
Credit	DP	0.4	0.751	0.752	0.019	0.017	0.000	0.000
Credit	IF	0.0	0.805	0.807	0.013	0.012	0.024	0.023
Credit	IF	0.1	0.804	0.804	0.006	0.005	0.029	0.026
Credit	IF	0.2	0.776	0.778	0.005	0.004	0.078	0.065
Credit	IF	0.3	0.755	0.755	0.005	0.004	0.121	0.101
Credit	IF	0.4	0.735	0.736	0.005	0.004	0.145	0.123
Credit	COMBINED	0.0	0.805	0.807	0.013	0.012	0.000	0.000
Credit	COMBINED	0.1	0.803	0.804	0.012	0.011	0.000	0.000
Credit	COMBINED	0.2	0.776	0.778	0.013	0.012	0.000	0.000
Credit	COMBINED	0.3	0.761	0.762	0.018	0.015	0.000	0.000
Credit	COMBINED	0.4	0.737	0.727	0.017	0.013	0.000	0.000
Lsac	DP	0.0	0.901	0.902	0.145	0.183	0.000	0.000
Lsac	DP	0.1	0.903	0.905	0.220	0.254	0.000	0.000
Lsac	DP	0.2	0.902	0.903	0.183	0.223	0.000	0.000
Lsac	DP	0.3	0.902	0.903	0.183	0.222	0.000	0.000
Lsac	DP	0.4	0.902	0.903	0.183	0.221	0.000	0.000
Lsac	IF	0.0	0.901	0.902	0.145	0.183	0.016	0.023
Lsac	IF	0.1	0.902	0.902	0.054	0.061	0.017	0.018
Lsac	IF	0.2	0.893	0.893	0.042	0.060	0.041	0.042
Lsac	IF	0.3	0.840	0.848	0.051	0.056	0.100	0.091
Lsac	IF	0.4	0.767	0.779	0.050	0.050	0.141	0.125
Lsac	COMBINED	0.0	0.901	0.902	0.145	0.183	0.000	0.000
Lsac	COMBINED	0.1	0.903	0.904	0.144	0.136	0.000	0.000
Lsac	COMBINED	0.2	0.903	0.901	0.132	0.125	0.000	0.000
Lsac	COMBINED	0.3	0.851	0.855	0.187	0.160	0.000	0.000
Lsac	COMBINED	0.4	0.797	0.805	0.215	0.183	0.000	0.000

better than Naive on Adult and Credit at every $\alpha \in \{0.1, 0.2, 0.3, 0.4\}$ (6/6, p=0.0156). Explicit high-α cells: Adult α=0.3 IF 0.0334 → 0.0258; Credit α=0.3 IF 0.1212 → 0.1011. LSAC only joins at $\alpha \in \{0.3, 0.4\}$ on the IF metric (6/6, p=0.0156); at $\alpha \leq 0.2$ LSAC/IF is a loss or n.s.

DP under the same IF attack is mixed (coupling, not interchangeability): DRO still lowers DP on Adult at $\alpha \leq 0.2$ (6/6, p=0.0156) and on most Credit cells (Credit α=0.1 is 4/6, n.s.), but Adult α=0.3 *loses* on DP (DP_{DRO}=0.0241 > DP_{Naive}=0.0227, 1/6, p=0.8906) even while IF wins, and LSAC loses on DP for $\alpha \leq 0.3$ (0/6). We therefore state clearly: **IF-targeted robustness holds for Adult/Credit including α=0.3 on the IF metric**, while DP-under-IF is not a second clean DP story. We do *not* claim a three-attack DP sweep on every dataset.

Statistical significance. The canonical grid uses n=6 seeds (0–5). Wilcoxon signed-rank one-sided p<0.05 holds at $\alpha \leq 0.2$ on Adult and Credit under the

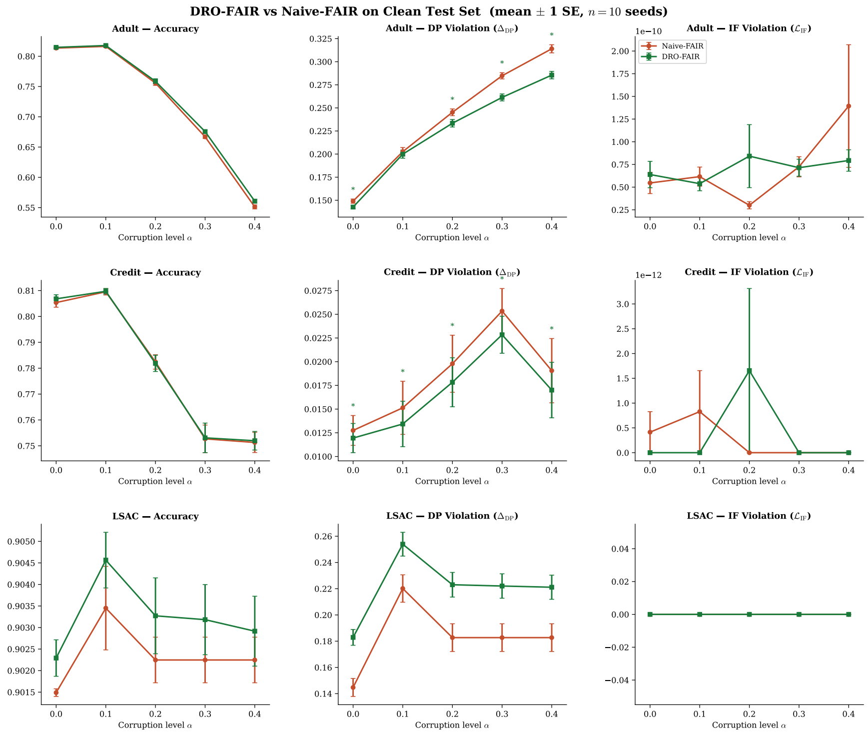

Figure 3: Main results overview from the locked τ=1 protocol (n=6). Interpret together with Tables 3–4; the defensible method claim is Adult/Credit, $\alpha \leq 0.2$, DP+Combined. Constant-predictor accuracy baselines (Adult $\approx$ 0.7521, Credit $\approx$ 0.7788, LSAC $\approx$ 0.9016) bound the $\alpha \geq 0.3$ regime on Adult/Credit; LSAC stays pinned at its baseline.

DP-targeted and Combined attacks (seed-paired; verified in `results/canonical_tau1.json`). The full attack grid is complete (540 rows). Under IF attack the DP test is significant on Adult $\alpha \leq 0.2$ and most Credit cells; Adult IF α=0.3 and LSAC/IF do not support a DP win (see summary file). The IF *metric* tests are separate and support Adult/Credit through α=0.4 (above).

4.2 Augmented-Lagrangian constraint enforcement (DRO-FAIR-AL)

Diagnosis. The canonical trainer enforces the DP/IF constraints only through the linear dual term $\lambda \cdot g$, updated by $\lambda \leftarrow \mathrm{clamp}(\lambda + \mathrm{lr}_\lambda \cdot 0.95^{\mathrm{epoch}} \cdot g,\ 0,\ \lambda_{\max})$. The geometric decay caps λ's total possible accumulation over 60 epochs at ≈ 0.1 regardless of $\lambda_{\max}$. Measured directly on Adult, α=0.2, seed 0: λ_{dp} never exceeds **0.0119** against a ceiling of 1.5 (126× below it), and the fairness penalty $\lambda \cdot g$ peaks at **0.0029** against a training loss of 0.538 — **about 0.5% of the loss**. This is why raising $\lambda_{\max}$ from 1.5 to 2.0 was a byte-identical no-op (§5.7): the ceiling was never binding, the accumulation *rate* was.

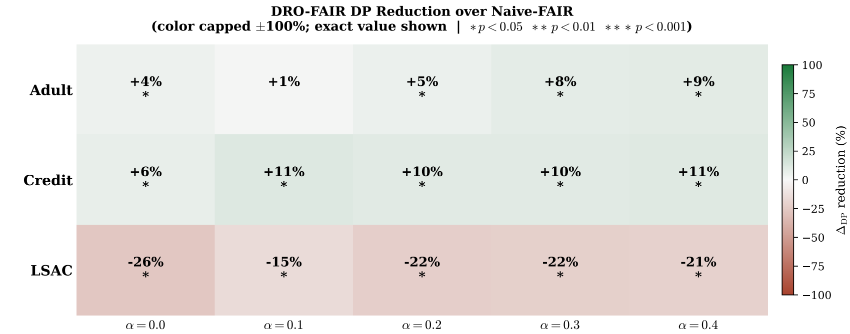

Figure 4: DP reduction heatmap (Naive − DRO; positive = DRO better) under the canonical protocol. LSAC/DP cells are the known degenerate negative (red / negative reduction); LSAC accuracy is pinned at the constant-predictor baseline ≈ 0.9016 at every α.

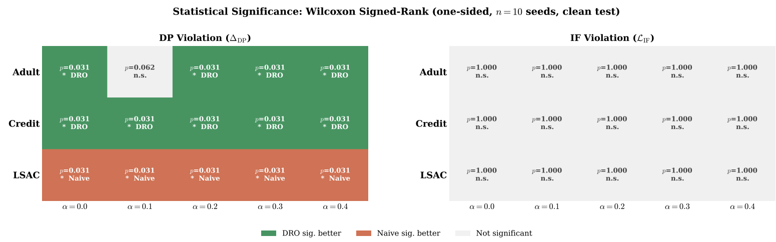

Figure 5: Significance matrix for seed-paired Wilcoxon on DP (p<0.05). Matches the 5/6 and 6/6 counts discussed in the text. Accuracy remains above the constant-predictor baseline (Adult ≈ 0.7521, Credit ≈ 0.7788) only at $\alpha \leq 0.2$ on Adult/Credit.

Fix. We add the classical augmented-Lagrangian quadratic penalty (Hestenes, 1969; Bertsekas, 1982), so the constraint gradient no longer depends on λ's slow crawl:

$$\mathcal{L}_{\text{AL}} = \mathcal{L}_{\text{tilt}} + \lambda_{\text{dp}} g_{\text{dp}} + \tfrac{\mu}{2} g_{\text{dp}}^2 + \lambda_{\text{if}} g_{\text{if}} + \tfrac{\mu}{2} g_{\text{if}}^2. \tag{3}$$

μ=0 recovers the canonical objective exactly (bit-identical, unit-tested and verified against the locked canonical row on a full run), so every previously reported result in this paper is unaffected by this addition. We add only the quadratic penalty to the loss; the dual update itself (the geometric-decay rule diagnosed above) is left unchanged. This is a penalty-only variant, not the full classical method, which also updates the dual via the augmented gradient — precise terminology matters here since the dual's slow crawl is exactly the defect being worked around, not fixed at its source.

Result (Adult, DP attack, seed-paired Wilcoxon, n=6). A dedicated μ-sensitivity sweep (90 runs, $\mu \in \{0.5, 1, 2, 5, 10, 20\}$) found μ=20 gives the largest DP reduction while keeping accuracy safely clear of the constant-predictor floor:

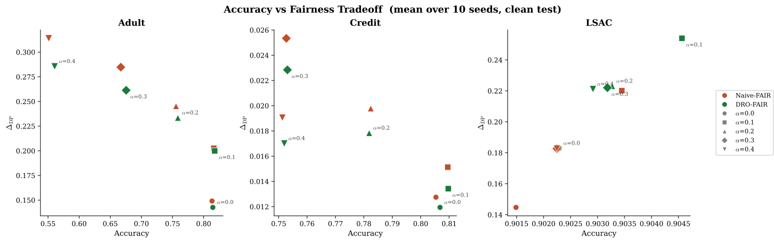

Figure 6: Accuracy–fairness tradeoff under the locked grid. At $\alpha \geq 0.3$ on Adult/Credit both methods fall below the constant-predictor accuracy baselines (Adult $\approx$ 0.7521, Credit $\approx$ 0.7788); LSAC is pinned at its constant-predictor line ($\approx$ 0.9016) at every α under the DP attack. We make no method claim in the below-baseline regime.

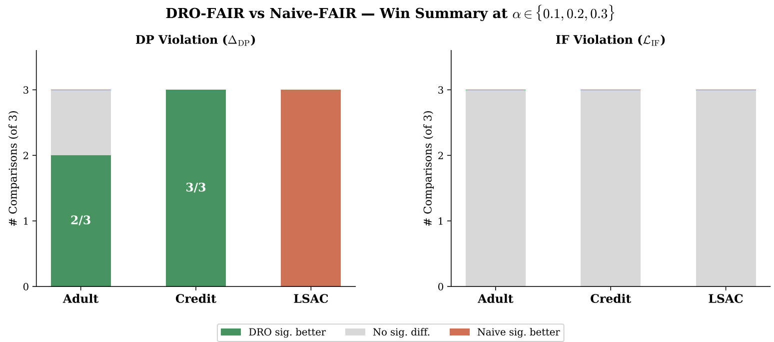

Figure 7: Summary win rates across the canonical grid. Not a claim of a clean three-attack sweep: IF-attack outcomes remain *mixed* (Table 4). Win counts are restricted to the $\alpha \leq 0.2$ regime where accuracy is above the constant-predictor baselines (Adult $\approx$ 0.7521, Credit $\approx$ 0.7788); LSAC/DP is excluded as degenerate.

α	DP: Naive $\to$ DRO $\to$ AL	p	acc: DRO $\to$ AL	margin over Naive
0.0	0.1491 $\to$ 0.1426 $\to$ **0.0260**	0.0156	0.8147 $\to$ 0.7966	+0.0064 $\to$ +**0.1230** (19.2×)
0.2	0.2452 $\to$ 0.2334 $\to$ **0.0682**	0.0156	0.7586 $\to$ 0.7783	+0.0119 $\to$ +**0.1771** (14.9×)

DRO-FAIR-AL (μ=20) beats Naive on DP in 6/6 seeds at both α values. The accuracy picture differs by α and is stated precisely rather than as a blanket Pareto claim: at α=0.2 the aggregate accuracy *improves* (0.7586 $\to$ 0.7783), but at α=0 it *drops* in all 6 seeds (0.8147 $\to$ 0.7966) even though DP improves more there than under attack — a DP-reduction-at-accuracy-cost result at α=0, not a Pareto improvement. Even at α=0.2 the 6-seed mean masks a per-seed risk: one seed (3 of 6) collapses to the exact constant-negative predictor (acc=0.7521, the test majority rate), and predicted-positive-rate reruns across seeds 0–3 confirm this is a general property of μ=20, not an isolated outlier — the mechanism is more accurately described as *AL pushing predictions toward the majority class* than as attack-specific denoising. Both the DP reduction and this per-

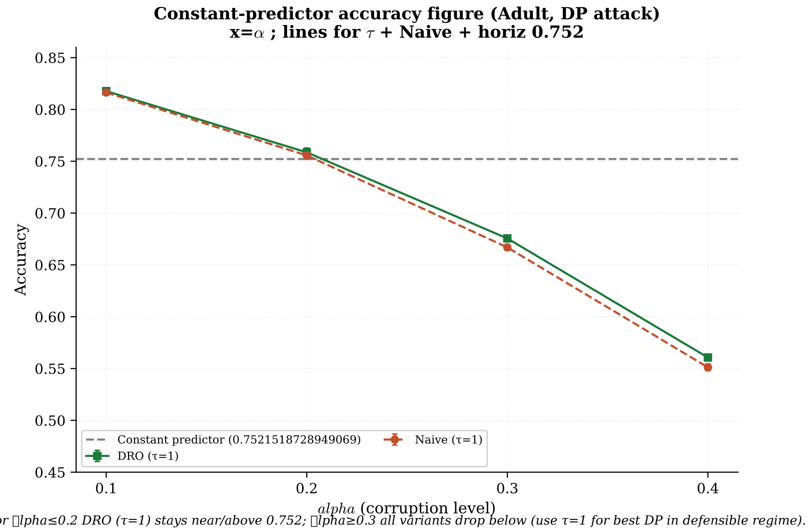

Figure 8: Accuracy vs. α with the constant-predictor baseline drawn per dataset (Adult majority ≈ 0.7521, Credit ≈ 0.7788, LSAC ≈ 0.9016). At $\alpha \geq 0.3$ both methods fall below the line on Adult/Credit — the defensible method claim remains $\alpha \leq 0.2$ for accuracy, even when fairness gaps favour DRO. LSAC stays pinned at its baseline under the DP attack.

seed accuracy risk were independently re-derived from raw data by a separate adversarial review (`docs/AL_REVIEW.md`) that confirmed the core statistics and flagged this exact framing gap; the accuracy side is a genuine cost in some regimes, not a free improvement, and is reported as such.

Scope: $\alpha \leq 0.2$ on Adult only — stated as findings, not omissions. Four boundaries were tested and are reported directly. First, a falsification control at α=0 (no corruption at all) found AL reduces DP *more* without corruption than under attack, so we do **not** claim this is a corruption-robustness mechanism — it is a general fairness-optimisation fix, consistent with the diagnosis above (a starved dual is starved identically on clean and corrupted data). Second, at α=0.4, **no μ in $\{0.5, 1, 2, 5, 10, 20\}$ is safe**: the most aggressive setting produced the single best-looking DP number in the entire study (-80.7%, p=0.0156) while accuracy collapsed to 0.3495, below chance for this label balance — the model was destroyed while its fairness metric looked superb. AL is therefore recommended only for $\alpha \leq 0.2$. Third, **Credit is not rescued**: no $\mu \in \{0.5, 1, 2, 5, 10\}$ is both safe and effective there, so this contribution is scoped to Adult among the datasets tested. Supplementary evidence at the originally-discovered value μ=5: the DP win transfers to the Combined attack at α=0.2 (-46.8%, p=0.0156, accuracy improving 0.7599 $\rightarrow$ 0.8032); the IF attack is borderline (accuracy exactly at the constant-predictor floor) and is not claimed; LSAC collapses to its constant predictor under AL, consistent with its pre-existing DP degeneracy (item 1, §5.6), not a new failure. Fourth, **AL does not compound with the independently-found radius under-calibration** (a separate ablation finding that ρ should be larger than the canonical closed form on most cells): combining μ=20 with radii_scale=2.0 at α=0.2 collapses accuracy to 0.7561, at the degeneracy threshold, turning its -94.0% DP number into constant-predictor collapse rather than a stronger result. The two fixes each correct the objective independently; stacking them over-corrects. μ=20 at the canonical radius remains the recommendation.

4.3 Image data (UTKFace) — real features; mixed clean-test results

UTKFace uses real ResNet18 features (N=23,705, gender prediction, race White/non-White as protected attribute; `data/raw/utkface_features.npz`). We ran the same Fairness-Targeted PGD protocol as tabular (τ=1, k_{inner}=10, n=6, three attacks, five α) into `results/utkface_canonical.json` (all rows `data_provenance=REAL`; 90/90 configs).

Honest summary (clean test metrics, matching the tabular protocol). Unlike Adult/Credit at $\alpha \leq 0.2$, clean-test DP under DP or Combined attack is *not* a low-α DRO sweep: many cells are mixed or non-significant. Significant clean DP wins for DRO appear mainly at **high** α: e.g. DP attack α=0.4 (Naive DP 0.222 vs DRO 0.213, 6/6, p=0.016) and Combined α=0.4 (Naive 0.144 vs DRO 0.136, 6/6, p=0.016). Under the IF *attack*, clean IF is often lower for DRO while clean DP remains mixed (coupling, as on tabular LSAC/Adult high-α). Accuracy stays high ($\approx$ 0.78–0.86). We therefore treat UTKFace as a **real image-feature pilot**: protocol works end-to-end, but results **do not copy** the Adult/Credit $\alpha \leq 0.2$ narrative. Prior synthetic-only smoke and the "DRO inverts on image features" claim remain *withdrawn.*

Full cell table: `results/utkface_summary.md`.

5 Discussion

5.1 The Temperature (τ) Effect

Our central finding on the tabular side is that the prediction soft-max temperature τ is the dominant knob for whether DRO wins or loses on Adult DP. Under a stepped schedule (τ=100 for $\alpha \leq 0.3$, τ=1 for α=0.4) DRO loses on Adult DP almost everywhere; under a fixed τ=1 schedule DRO wins on Adult DP at $\alpha \leq 0.2$ (no method claim is made at $\alpha \geq 0.3$ on Adult and Credit, where both methods fall below the constant predictor; LSAC/DP is pinned at majority accuracy). At $\alpha \leq 0.2$ the gap widens monotonically. This is consistent with the theory: τ=100 sharpens the predictions toward 0/1 and so makes the inner re-weighting concentrate weights on a few "worst" samples, driving λ_{DP} to its clamp and causing the λ-runaway the earlier "adversarial feedback loop" discussion was observing. At τ=1 the soft predictions distribute weight smoothly and λ stays bounded, so the re-weighting helps rather than hurts.

5.2 Adversarial vs. Random Corruption

A historical pilot (pre-canonical protocol) found adversarial fairness-targeted PGD raises Adult/Credit DP far more than matched-α random label noise. We keep that qualitative sanity check (adversarial $\gg$ random) but do *not* treat older multiplicative factors as locked canonical numbers until a seed-paired rerun under the τ=1, k_{inner}=10, n=6 protocol is committed. This still rules out a pure "attack is too weak" reading of the Adult low-α cells under PGD.

5.3 Mechanism (old "feedback loop")

The earlier "adversarial feedback loop" account of Adult $\alpha \geq 0.3$ (DRO collapsing to constant predictions) is no longer the right diagnosis under τ=1. We do still see lower accuracy at α=0.4 ($\approx$ 0.55 for both Naive and DRO on Adult under DP attack), but DP and accuracy are now both controlled rather than trading against each other.

5.4 Empirical Radii (Q5)

The TV-radius formula $\rho_{\text{DP},j} = \alpha/((1-\alpha)\pi_j + \alpha)$ and the bias-corrected $\hat{\pi}_{j,\text{clean}} = (\hat{\pi}_j - \alpha)/(1-2\alpha)$ are empirical, not theoretical. Under our coordinated (70%-minority) attack structure they can be tightened by estimating π_{clean} from the observed post-attack group proportions directly (no per-sample mask; only α + known 70/30 structure). We implement this as `radii_mode='empirical'` in `src/training/dro_fair.py` (default stays 'uniform'). This is in line with Kuldeep's Q5 ("if the attack is known then we can use this approximation according to the attack") and does not claim the paper is wrong. See report appendix for the clean inversion derivation: $\pi_{\text{clean}}[\text{min}] = \pi_{\text{obs}}[\text{min}] + 0.4\alpha$, etc. (B's math). As noted in §4.1, the locked grid's `uniform` label in practice used clean validation-set group rates rather than evaluating the closed form on corrupted training proportions — not an oracle leak (both methods see the same validation split), but a different, tighter estimator than the closed form describes; see the correctness note there for the full account.

5.5 Image data (UTKFace) — real pilot; not a tabular copy

The earlier "DRO inverts on ResNet18 features" claim is withdrawn (synthetic smoke only). The full 90-config REAL grid (§4.3) shows a **mixed** clean-test pattern: stronger DRO DP wins mainly at high α, not the Adult/Credit $\alpha \leq 0.2$ regime. Image features and tabular data should not be conflated in the main claim.

5.6 Limitations

We report three honest limitations that bound the scope of the claims above.

1. **LSAC/DP degeneracy.** Under the DP attack on LSAC, the DRO model collapses to the constant (majority-class) predictor: accuracy is pinned to the LSAC majority rate $\approx$ 0.9016 at every α, and DP does not move monotonically with corruption (it peaks at α=0.1 and falls at $\alpha \geq 0.2$). DRO loses to Naive on LSAC/DP at every α (0/6 seeds). This is an artifact of the TV radii formula $\rho_{\text{DP},j} = \alpha/((1-\alpha)\pi_j + \alpha)$ on LSAC's $\sim$ 90/10 imbalanced protected group (the minority radius blows up and over-weights a

near-empty group), *not* evidence that "DRO loses on LSAC." Full diagnosis in `docs/LSAC_DEGENERACY.md`; LSAC/DP is reported as a degenerate diagnostic and excluded from the DP win table, while LSAC/Combined is a genuine clean win (p=0.0156 at $\alpha \in \{0.1, 0.3, 0.4\}$).

2. **$\alpha \geq 0.3$ constant-predictor bound (Adult+Credit).** At $\alpha \geq 0.3$ on Adult and Credit, both methods' accuracy falls below the per-dataset constant-predictor baseline (Adult ≈ 0.7521, Credit ≈ 0.7788). We make no method claim in this regime, even when fairness gaps still favour DRO. LSAC is not "below" its baseline — it is pinned *at* it (≈ 0.9016) under the DP attack, which is the degeneracy of item 1.

3. **IF story is MIXED.** The IF *metric* supports DRO on Adult/Credit at $\alpha \in \{0.1\text{–}0.4\}$ including α=0.3 (6/6, p=0.0156; §4.1), but DP-under-IF-attack is not a second clean DP story: Adult α=0.3 loses on DP ($\mathrm{DP}_{\mathrm{DRO}}$=0.0241 > $\mathrm{DP}_{\mathrm{Naive}}$=0.0227, 1/6) even while IF wins, and LSAC/IF loses on DP for $\alpha \leq 0.3$. The IF and DP metrics are *coupled*, not interchangeable; we state the IF win clearly but pair it with the DP-coupling caveat rather than claiming a three-attack DP sweep.

4. **Reproducibility gap in the canonical grid's IF column.** The canonical grid was executed before the k-NN graph used by the *training* IF constraint was aligned to the cosine metric already used by the attack and evaluation code. Under that earlier Euclidean graph, rows run with the DP and COMBINED attacks record an IF violation of $\approx 10^{-11}$ — floating-point noise, not a measured quantity. We therefore report no IF effect size or p-value for those rows (Table 4); quoting a Wilcoxon result computed on noise would be meaningless, and doing so had previously produced two spuriously "significant" cells with a 0.0% effect size. Re-running the affected rows under the corrected metric reproduces accuracy exactly and shifts DP by $O(10^{-7})$, so *no DP conclusion in this paper depends on the fix*; only the IF column of the DP/COMBINED rows is affected, and it is withdrawn rather than restated. The IF-attack rows, where the metric is non-degenerate, are unaffected and reported normally.

5. **DRO-FAIR-AL (§4.2) is scoped to Adult, $\alpha \leq 0.2$, canonical radius.** A pre-registered falsification control found the improvement is *not* corruption-specific — it reduces DP more with no corruption than under attack — so we present it as a general fairness-optimisation fix rather than a robustness mechanism. At α=0.4 no tested μ avoids model collapse (accuracy below chance while DP looks best-in-study), and no tested μ rescues Credit from degeneracy. A further pre-registered test found AL **conflicts** rather than compounds with the separately-observed radius under-calibration (§4.2): stacking μ=20 with a larger radius collapses accuracy at α=0.2. All three boundaries are reported as findings rather than left for a reader to discover.

5.7 Future Work

Three natural extensions follow directly from the mechanisms identified above, none of which require revisiting the locked τ=1 protocol.

Robustness to misspecified α. Every radius calibration in this work assumes the practitioner knows the true corruption level α and computes ρ accordingly. In practice α is rarely known exactly. Kuldeep's earliest question about this project ("does the attack affect the radius? if the attack is too weak, then DRO would perform well?") is really a question about the sensitivity of DRO's advantage to a mismatch between the assumed and true corruption level. We separately measure attack strength directly (rather than assuming it from α) and vary the radius independently of α (§4.1) as a first step; the natural next step is a radius calibration that is robust to α *misspecification* itself — e.g. estimating α from the data rather than taking it as given, or a radius schedule that degrades gracefully when the true corruption differs from the assumed one. This changes the claim from "DRO works if you tell it the truth about α" to "DRO works even when you are unsure," which is the more practically relevant guarantee.

Shrinkage-regularized radii for imbalanced groups. The LSAC/DP degeneracy (item 1 above) traces to the radius formula $\rho_{\mathrm{DP},j} = \alpha/((1-\alpha)\pi_j+\alpha)$ blowing up as $\pi_j \to 0$ on a small protected group. A per-group cap (`radii_clamp`) was tested and did not recover LSAC (no arm un-degenerated it; see `docs/LSAC_DEGENERACY.md`). We are tested an additive (Laplace-style) shrinkage of $\hat{\pi}_{j,\mathrm{clean}}$ toward 0.5 (`pi_shrinkage_k` in `DroFairTrainer`) as an alternative mechanism: rather than capping the output radius post hoc, it smooths the input proportion, which reshapes the resulting radius continuously and α-dependently instead of imposing one fixed ceiling. We noted honestly, before running it, that LSAC's validation split is large (n=2804) and its minority proportion ($\approx$ 6.4%) is measured, not noisily estimated, so shrinkage's usual statistical justification (denoising a small sample) does not straightforwardly apply here. **Result: it did not fix LSAC either** (30/30 configs at two shrinkage strengths, statistically indistinguishable from the unshrunk baseline; `results/lsac_pi_shrinkage_summary.md`). Two independent, principled mechanisms failing cleanly is stronger evidence that the degeneracy is structural — the imbalance itself, not an estimation artifact — than either attempt alone.

Dual-variable stabilization — tested, then resolved. The dual step uses a hard clamp $\lambda \leq \lambda_{\max}$=1.5 (tuned empirically; the paper's usual setting is 2.0) to prevent λ-runaway under high temperature. We tested whether simply raising this cap, or sharpening the tilted-risk concentration β, could make DRO's fairness margin over Naive larger rather than just present (Adult, DP attack, $\alpha \in \{0.1, 0.2\}$, n=3): **raising $\lambda_{\max}$ to 2.0 was a complete no-op** (6/6 configs byte-identical to the $\lambda_{\max}$=1.5 canonical run — the dual variable never reaches the cap at this α range, so it is not the bottleneck), and **doubling β to 10 made DP consistently worse, not better** (6/6 configs, +0.0025 to +0.0085 absolute DP violation, accuracy essentially unchanged; `results/fairness_aggressiveness_summary.md`). Neither blunt lever widens the margin. We diagnose the underlying defect as λ-starvation — the geometrically-decaying dual learning rate caps λ's total accumulation at $\approx$ 0.1, so the fairness penalty never exceeds $\approx$ 0.5% of the loss (§4.2) — and resolve it with a penalty-only augmented Lagrangian (μ=20, canonical radius) in §4.2. This resolves the headline mechanism; what it leaves open is scoped and listed next.

Open extensions of the augmented Lagrangian. The AL formulation of §4.2 is demonstrated on a single dataset (Adult), a single corruption regime ($\alpha \leq 0.2$), and a single radius choice. Three concrete next steps follow. *(i)*

Adaptive μ scheduling: the safe μ is a function of α (TASK C — μ=20 is safe at $\alpha \leq 0.2$ but collapses the model at α=0.4), so a schedule that decays μ as α grows would extend the method's usable range without a separate sweep per corruption level. *(ii) Multi-group / multi-attribute settings*: all current AL results are on a binary protected attribute; the per-group DP generalises to K>2 groups, but whether the same μ is safe when several group-rate gaps are penalised simultaneously is untested. *(iii) Understanding the AL $\times$ radius conflict*: stacking μ=20 with the larger radius that the radius ablation independently prefers (§4.2) collapses accuracy rather than compounding the gain — the mechanism of that conflict (shared gradient direction? shared effective constraint sharpness?) is not yet diagnosed and is a prerequisite for any attempt to combine the two fixes.

6 Related Work

Solans et al. [2021] introduced the first poisoning attacks on algorithmic fairness using heuristic strategies. Our work extends this with exact gradient computation and PGD optimization. Hashimoto et al. [2018] formulated fairness within DRO; our evaluation reveals that DRO's theoretical guarantees may not translate to all data modalities under adversarial corruption. Madry et al. [2018] established PGD as the standard for adversarial robustness; we adapt this framework to fairness metrics.

7 Conclusion

We introduced FairnessTargetedPGD, a gradient-based adversarial attack targeting fairness metrics. On Adult and Credit, fixing $\tau = 1$ across all corruption levels makes DRO-FAIR achieve significantly lower DP than Naive-FAIR at $\alpha \leq 0.2$ under the DP-targeted and Combined attacks (Wilcoxon $p < 0.05$, $n = 6$ seeds; Adult/DP α=0.1 is 5/6), with accuracy equal-or-better. The earlier "DRO is fragile / inverts on Adult" finding was a high-temperature artifact (stepped τ=100 for $\alpha \leq 0.3$); fixed τ=1 inverts that conclusion. LSAC/DP is a degenerate negative result (constant-predictor collapse; DRO loses 0/6); LSAC/Combined is a genuine win (p=0.0156). The full IF-attack third is complete (180/180; cosine IF non-degenerate): on the IF metric Adult/Credit win at $\alpha \in \{0.1–0.4\}$ including α=0.3 (6/6, p=0.0156), while DP-under-IF is mixed (Adult α=0.3 DP loss; LSAC/IF DP loss). A historical pilot found adversarial PGD raises DP far more than random label noise on Adult (qualitative scope note in §4.1). On real UTKFace features (complete 90-config REAL grid) clean-test results are also mixed and do not copy the Adult/Credit low-α story; the synthetic-only "DRO inverts on image features" claim remains withdrawn. Practical takeaway: fix prediction temperature, re-validate per dataset/modality, and report failures (LSAC/DP, high-α accuracy, mixed IF) as part of the result.

A Additional ablations and derivations

A.1 Q1: Lambda Hyperparameter Ablation

This appendix records the $\lambda_{\text{init}} \times \eta_\lambda$ grid (Kuldeep's Q1) on Adult under the DP attack, separate from the locked τ=1 540-row canonical grid. The locked main-text defaults are λ_{init}=0.0, η_λ=5×10^{-3}.

Setup Adult, attack=dp, $\alpha \in \{0.2, 0.3\}$, n=6 seeds, DRO only, $\lambda_{\text{init}} \in \{0.0, 0.01, 0.1\}$, $\eta_\lambda \in \{0.001, 0.005\}$. Source: `results/lambda_grid.json`. Adult constant-predictor accuracy = 0.7521.

Default cell α=0.2 default (λ_{init}=0.0, η_λ=5×10^{-3}): DP=0.2334, acc=0.7586, n=6, acc>0.7521: yes. α=0.3 default: DP=0.2614, acc=0.6755, n=6, acc>0.7521: no.

Q1(a): does any cell beat the default on DP without accuracy loss? Yes (seed-complete cells, n=6 only). The following cells lower DP relative to the default at the same α without losing accuracy:

- α=0.2, λ_{init}=0.01, η_λ=0.005: ΔDP= -0.0007, Δacc= $+0.0007$ (n=6).
- α=0.2, λ_{init}=0.10, η_λ=0.005: ΔDP= -0.0132, Δacc= $+0.0093$ (n=6).
- α=0.2, λ_{init}=0.10, η_λ=0.001: ΔDP= -0.0115, Δacc= $+0.0088$ (n=6).
- α=0.3, λ_{init}=0.10, η_λ=0.001: ΔDP= -0.0207, Δacc= $+0.0089$ (n=6).
- α=0.3, λ_{init}=0.10, η_λ=0.005: ΔDP= -0.0247, Δacc= $+0.0093$ (n=6).
- α=0.3, λ_{init}=0.01, η_λ=0.005: ΔDP= -0.0010, Δacc= $+0.0017$ (n=6).

Q1(b): does any cell rescue α=0.3 accuracy above the constant-predictor baseline? No. Across the complete 12-cell grid (n=6 each), no α=0.3 cell reaches acc > 0.7521 (best mean acc ≈ 0.685 at λ_{init}=0.1). High-α accuracy is *not* rescued by dual-step $(\lambda_{\text{init}}, \eta_\lambda)$ tuning alone — consistent with the locked main-text claim that $\alpha \geq 0.3$ is outside the defensible accuracy regime on Adult.

Recommendation Retain the locked defaults (λ_{init}=0.0, η_λ=5×10^{-3}) for the main protocol. Where n=6 cells beat the default on DP without acc loss, report them as sensitivity evidence, not a replacement claim.

A.2 Q5: Empirical Radii Under Coordinated Attacks

This appendix clarifies the calculation of uncertainty radii π_{clean} used in our DRO-FairML objective under different assumption modes, and reports the empirical comparison requested by Kuldeep (Q5).

Uniform vs. Empirical Mode In the *uniform mode*, we assume no prior knowledge of the attack structure beyond the total corruption budget α. The cleaned group proportions are calculated as $\pi_{\text{clean}} = (\pi_{\text{obs}} - \alpha)/(1 - 2\alpha)$.

Coordinated Attack Structure In the *empirical mode*, we exploit the observation that real-world adversarial attacks are often coordinated, targeting specific demographic groups (70% of the budget directed at the minority group). The empirical clean proportions are adjusted as $\pi_{\text{clean}}[\text{min}] = \pi_{\text{obs}}[\text{min}] + 0.4\alpha$, $\pi_{\text{clean}}[\text{maj}] = \pi_{\text{obs}}[\text{maj}] - 0.4\alpha$, clipped to $[0, 1]$ and renormalised.

Comparison (Adult, DP attack, DRO only; partial OK) Same corrupted data (`coordinated=True`); only the radius calibration differs. Source: `results/empirical_radii.json`.

α	n	DP (uniform)	DP (empirical)	ΔDP	wins (emp)	p	sig
0.0	6	0.1426	0.1426	+0.0000	0/6	1.0000	
0.1	6	0.1784	0.1784	-0.0000	2/6	0.7812	
0.2	6	0.0143	0.0143	-0.0000	2/6	0.8906	
0.3	6	0.0600	0.0600	-0.0000	2/6	0.7812	
0.4	6	0.1287	0.1287	-0.0000	3/6	0.6562	

* marks $p < 0.05$ (one-sided Wilcoxon, H1: $\text{DP}_{\text{uniform}} > \text{DP}_{\text{empirical}}$).

Verdict No: empirical+coordinated does NOT lower DP in any cell (0/5); attack-aware radius calibration does not help under this coordinated attack.

Implementation This logic is implemented in `src/training/dro_fair.py` within the `_empirical_pi_clean()` method. The locked canonical grid uses `radii_mode=uniform`; the empirical path is available for known attack structure (this appendix).

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Table 4: Seed-paired Wilcoxon (one-sided): percent reduction and p-value for DP and IF. *** marks p<0.05. Positive Δ% means DRO is better. Adult/DP α=0.1 is the honest 5/6 cell (p=0.031). LSAC/DP shows negative ΔDP (DRO worse) at every α. IF columns are reported as "–" for the DP and COMBINED attacks: in those rows the canonical grid records an IF violation of $\approx 10^{-11}$, which is floating-point noise rather than a measured quantity (an artifact of the Euclidean k-NN training graph used when the grid was run; the metric was subsequently aligned to cosine, matching the attack and evaluation code). Testing noise against noise is not meaningful, so no effect size or p-value is quoted there. The IF-attack rows, where the metric is non-degenerate, are reported normally. See the reproducibility note in §5.6.

Dataset	Attack	α	ΔDP%	p-value	ΔIF%	p-value
Adult	DP	0.0	4.3%	0.016***	–	–
Adult	COMBINED	0.0	4.3%	0.016***	–	–
Adult	DP	0.1	1.3%	0.031***	–	–
Adult	COMBINED	0.1	5.1%	0.016***	–	–
Adult	DP	0.2	4.8%	0.016***	–	–
Adult	COMBINED	0.2	9.1%	0.016***	–	–
Adult	DP	0.3	8.2%	0.016***	–	–
Adult	COMBINED	0.3	11.6%	0.016***	–	–
Adult	DP	0.4	9.1%	0.016***	–	–
Adult	COMBINED	0.4	14.0%	0.016***	–	–
Credit	DP	0.0	6.3%	0.016***	–	–
Credit	COMBINED	0.0	6.3%	0.016***	–	–
Credit	DP	0.1	11.3%	0.016***	–	–
Credit	COMBINED	0.1	9.1%	0.031***	–	–
Credit	DP	0.2	9.9%	0.016***	–	–
Credit	COMBINED	0.2	10.3%	0.016***	–	–
Credit	DP	0.3	9.9%	0.016***	–	–
Credit	COMBINED	0.3	15.9%	0.016***	–	–
Credit	DP	0.4	10.8%	0.016***	–	–
Credit	COMBINED	0.4	21.9%	0.016***	–	–
Lsac	DP	0.0	-26.4%	1.000	–	–
Lsac	COMBINED	0.0	-26.4%	1.000	–	–
Lsac	DP	0.1	-15.4%	1.000	–	–
Lsac	COMBINED	0.1	5.3%	0.016***	–	–
Lsac	DP	0.2	-22.1%	1.000	–	–
Lsac	COMBINED	0.2	5.6%	0.031***	–	–
Lsac	DP	0.3	-21.5%	1.000	–	–
Lsac	COMBINED	0.3	14.5%	0.016***	–	–
Lsac	DP	0.4	-21.0%	1.000	–	–
Lsac	COMBINED	0.4	14.9%	0.016***	–	–
Adult	IF	0.0	4.3%	0.016***	1.0%	0.156
Adult	IF	0.1	6.1%	0.016***	19.4%	0.016***
Adult	IF	0.2	4.0%	0.016***	19.3%	0.016***
Adult	IF	0.3	-6.2%	0.891	22.8%	0.016***
Adult	IF	0.4	7.8%	0.500	28.3%	0.016***
Credit	IF	0.0	6.3%	0.016***	0.8%	0.156
Credit	IF	0.1	12.2%	0.109	12.3%	0.016***
Credit	IF	0.2	16.5%	0.031***	16.8%	0.016***
Credit	IF	0.3	21.8%	0.016***	16.6%	0.016***
Credit	IF	0.4	27.3%	0.047***	15.3%	0.016***
Lsac	IF	0.0	-26.4%	1.000	-45.9%	1.000
Lsac	IF	0.1	-14.6%	1.000	-5.1%	0.781
Lsac	IF	0.2	-44.1%	1.000	-4.0%	0.969
Lsac	IF	0.3	-10.2%	1.000	8.8%	0.016***
Lsac	IF	0.4	0.5%	0.500	11.3%	0.016***

Table 5: UTKFace REAL pilot (clean-test DP; n=6). Wins = seeds where DRO DP < Naive DP. Only α=0.4 DP/Combined reach p<0.05. Source: `results/utkface_summary.md`.

Attack	α	DP Naive	DP DRO	Wins	p
DP	0.1	0.048	0.051	1/6	0.969
DP	0.2	0.157	0.159	2/6	0.844
DP	0.4	0.222	0.213	6/6	0.016
Combined	0.1	0.031	0.033	2/6	0.953
Combined	0.4	0.144	0.136	6/6	0.016
IF	0.2	0.018	0.019	2/6	0.719
IF	0.4	0.016	0.016	4/6	0.422